PCCST501 – COMPUTER NETWORKS

Assignment 3 | Module 3 – Data-Link Layer, Wireless LANs & Mobile IP

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Answers for Questions 1, 2 and 4–10 (Question 3 excluded as requested)

1. Responsibilities of the Data-Link Layer; LLC vs MAC

The Data-Link Layer is the second layer of the OSI model. It provides node-to-node delivery of data between directly connected devices.

• **Framing:** Divides network-layer packets into frames.

• **Physical addressing:** Uses MAC addresses to identify source and destination devices.

• **Error control:** Detects errors during transmission.

• **Flow control:** Controls the rate of data transmission between devices.

• **Access control:** Decides which device can use a shared communication medium.

Difference between LLC and MAC

LLC (Logical Link Control)	MAC (Media Access Control)
Provides an interface to the network layer.	Controls access to the transmission medium.
Helps identify the network-layer protocol.	Handles MAC addressing.
Supports logical link management.	Handles frame transmission over the medium.

Conclusion: The Data-Link Layer provides organized communication between directly connected devices.

2. Ethernet Frame Format and Purpose of Fields

An Ethernet frame is the unit of data transmitted over an Ethernet network. It contains addressing information, data and error-detection information.

Field	Size / purpose
Preamble	7 bytes; helps synchronize transmission.
SFD	1 byte; indicates the beginning of the frame.
Destination MAC	6 bytes; identifies the receiving device.
Source MAC	6 bytes; identifies the sending device.
Type/Length	2 bytes; identifies the upper-layer protocol or indicates data length.
Data and Padding	46–1500 bytes in standard Ethernet.
FCS	4 bytes; detects errors using a CRC value.

Conclusion: Ethernet frames support addressing, data transmission and error detection.

4. Switch Operation Using MAC Address Learning

An Ethernet switch connects multiple devices in a LAN. It forwards frames to the correct destination using MAC addresses.

• **Learning:** Reads the source MAC address and stores it with the incoming port.

• **Forwarding:** Checks the destination MAC address in its table.

• **Filtering:** Does not forward a frame to other ports if the destination is on the incoming port.

• **Flooding:** Sends a frame to all ports except the incoming port if the destination is unknown.

• **Updating:** Updates its table as frames arrive from different devices.

Example forwarding table

MAC address	Port
AA:AA:AA:AA:AA:01	Port 1
BB:BB:BB:BB:BB:02	Port 2
CC:CC:CC:CC:CC:03	Port 3

When a frame arrives for PC2, the switch checks the table and forwards the frame only through Port 2.

Conclusion: MAC address learning helps a switch forward frames efficiently and reduces unnecessary traffic.

5. Compare Hub, Switch and Bridge

Feature	Hub	Switch	Bridge
OSI layer	Physical	Data-Link	Data-Link
Forwarding	Sends data to all ports	Sends frames to required port	Forwards based on MAC addresses
Collision domains	One shared domain	Each port is a separate domain	Separates domains between segments
MAC learning	No	Yes	Yes
Performance	Low	High	Better than hub
Practical use	Older simple networks	Modern LANs	Connects/separates LAN segments

Explanation: A hub repeats signals to all devices. A switch learns MAC addresses and forwards frames to the correct port. A bridge connects LAN segments and filters frames using MAC addresses.

Conclusion: Switches are commonly used in modern LANs because they provide efficient forwarding and separate collision domains.

6. CSMA/CD and Its Reduced Importance in Modern Ethernet

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It controls access to a shared Ethernet medium.

• **Carrier Sense:** A device checks whether the medium is free.

• **Multiple Access:** Multiple devices share the same medium.

• **Transmission:** If the medium is free, the device sends data.

• **Collision Detection:** The device checks whether a collision occurs.

• **Backoff:** If a collision occurs, the device stops, waits for a random time and retries.

Why is its importance reduced?

• Switches provide separate links for devices.

• Full-duplex communication allows sending and receiving simultaneously.

• Collisions do not normally occur on full-duplex Ethernet links.

• Therefore, CSMA/CD is not needed on modern switched full-duplex Ethernet.

Conclusion: CSMA/CD was important in shared Ethernet networks but is unnecessary on modern full-duplex switched links.

7. CSMA/CA in IEEE 802.11 and Comparison with CSMA/CD

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is used in IEEE 802.11 wireless LANs to reduce the possibility of collisions.

• The wireless device checks whether the channel is free.

• If the channel is busy, it waits.

• If free, it waits for a specified interval and selects a random backoff time.

• When the backoff reaches zero, it transmits data.

• The receiver sends an acknowledgment (ACK) if the frame is received successfully.

• If the ACK is not received, the sender may retransmit the frame.

• RTS/CTS may be used to reduce collisions in suitable situations.

CSMA/CA	CSMA/CD
Collision Avoidance	Collision Detection
Used in wireless LANs (IEEE 802.11).	Traditionally used in shared, half-duplex Ethernet.
Tries to prevent collisions before transmission.	Detects collisions during transmission.
Uses waiting and random backoff.	Stops after collision and retries later.
Wireless devices cannot reliably detect collisions while transmitting.	Devices on a shared wired medium can detect collisions.

Conclusion: CSMA/CA tries to avoid collisions in wireless networks, while CSMA/CD detects collisions in shared wired networks.

8. Design a Small Wireless LAN for a College Laboratory

A Wireless LAN (WLAN) connects computers and other devices using Wi-Fi without requiring a separate cable for every device.

Network design

Internet connection → **Router** → **Wireless Access Point (AP)** → **Computers and laptops**

Access point placement

• Place the AP in a central location and at a suitable height.

• Keep it away from thick walls and sources of interference.

• Use additional APs if one AP cannot cover the entire lab.

SSID

SSID means Service Set Identifier. It is the name of the wireless network. Example: **College_Lab_WiFi**.

Channel considerations

• Use non-overlapping channels to reduce interference.

• In 2.4 GHz, channels 1, 6 and 11 are commonly used non-overlapping choices.

• Use 5 GHz when suitable for better capacity and reduced interference.

• Select channels according to nearby networks and AP capabilities.

IP addressing

Parameter	Example
Network address	192.168.1.0/24
Router address	192.168.1.1
DHCP range	192.168.1.100–192.168.1.200
Subnet mask	255.255.255.0

Use DHCP to assign IP addresses automatically. Adjust these example values to avoid conflicts with existing networks.

Basic security requirements

• Use WPA2-AES or WPA3 security.

• Set a strong Wi-Fi password and change default administrator credentials.

• Provide a separate guest network if needed.

• Update AP firmware and restrict access to authorized users.

Conclusion: A properly designed WLAN provides reliable, secure and convenient wireless connectivity for a college laboratory.

9. Mobile IP and the Need for Mobility Support

Mobile IP is a network-layer mechanism that allows a mobile device to move from one network to another while maintaining its permanent IP address and ongoing communication.

Need for Mobile IP

When a device moves to another network, its network attachment changes. Without mobility support, communication may be interrupted because its IP address and network location are linked. Mobile IP helps the device remain reachable.

Important terms

• **Mobile Node:** The device that moves between networks.

• **Home Agent:** Maintains information about the mobile node's current location.

• **Foreign Agent:** In Mobile IPv4, assists a visiting mobile node in the foreign network.

• **Care-of Address:** An address representing the mobile node's current location.

• **Correspondent Node:** The device communicating with the mobile node.

Three phases of Mobile IP

• **1. Agent Discovery:** The mobile node discovers mobility agents and obtains a care-of address.

• **2. Registration:** The mobile node registers its care-of address with the home agent.

• **3. Data Transfer:** The home agent forwards packets towards the mobile node, typically using tunneling.

Working

• The mobile node starts in its home network.

• When it moves to another network, it discovers a care-of address.

• It registers this address with its home agent.

• The home agent tunnels packets towards the mobile node's current location.

• The mobile node receives packets and continues communication.

Conclusion: Mobile IP supports communication while a device moves between networks by maintaining its permanent identity and tracking its current location.

10. Compare a Wired Ethernet Frame and a Wireless LAN Frame

Ethernet and wireless LANs both use frames to transfer data, but their frame formats and access methods are different.

Feature	Wired Ethernet	Wireless LAN (IEEE 802.11)
Standard	IEEE 802.3	IEEE 802.11
Address fields	Destination and source MAC	Usually 3 or 4 MAC address fields, depending on frame settings
Header size	Typically 14 bytes	Typically 24 bytes for a basic three-address data frame
Error checking	FCS (4 bytes)	FCS (4 bytes)
Medium access	Switched full-duplex sends/receives simultaneously	Uses CSMA/CA to reduce collisions
Extra fields	Type/Length	Frame Control, Duration/ID, Sequence Control and address fields
Medium	Copper or fibre	Radio waves

Example: wired Ethernet frame

Suppose a computer sends an IP packet through an Ethernet switch.

Field	Illustrative example
Destination MAC	AA:BB:CC:DD:EE:02
Source MAC	AA:BB:CC:DD:EE:01
Type	0x0800 (IPv4)
Data	IP packet
FCS	Error-checking value

The switch uses the destination MAC address to forward the frame to the correct port.

Example: wireless LAN frame

Field	Purpose / example
Frame Control	Identifies frame type and control information
Address 1	Receiver address, commonly the AP's MAC address
Address 2	Transmitter address, commonly the laptop's MAC address
Address 3	Additional address, often the destination on the distribution system
Sequence Control	Contains sequence and fragment information
Data	IP packet
FCS	Error-checking value

The exact address meanings depend on transmission direction and the To DS/From DS bits.

Packet-level comparison using Wireshark

• Select an Ethernet frame and expand the Ethernet II section.

• Note the source MAC, destination MAC, type and frame length.

• Select a wireless frame and expand the IEEE 802.11 section.

• Note the frame type, transmitter and receiver addresses, sequence information and frame length.

• Compare the fields and observe how each frame supports delivery over its medium.

Note: These are illustrative examples, not actual Wireshark captures. Use real capture values if screenshots are required.

Conclusion: Ethernet frames support wired communication, while IEEE 802.11 frames include additional control and addressing information suited to wireless communication.

References

• IEEE 802.3 – Ethernet Standard.

• IEEE 802.11 – Wireless LAN Standard.

• IEEE 802.1D – MAC Bridges and Bridging.

• RFC 3344 – IP Mobility Support for IPv4.

• Wireshark User's Guide – Packet Analysis.